The Clay Navier–Stokes Problem:

Proposal of Reformulation and Solution

Claes Johnson[1]

[1]KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden

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written up with AI assistance (Claude, Anthropic).

1 Reformulation of the Navier–Stokes problem

On 8 September 2026 OpenAI presented an AI solution [22] to the official formulation of the Clay Navier–Stokes Millennium problem, as set out by Charles Fefferman [26]. We give evidence here that the official formulation does not reach the *turbulent* flows which are what incompressible flow at high Reynolds number in fact consists of, and that a construction can therefore satisfy it without bearing on turbulence — and so without bearing on anything essential to fluid flow.

What follows is, in substance, not new. Both the reformulation and the solution are essentially the content of Chapter 14 of [30], published in 2007, set out there as part of a computational programme rather than as a reply to the Clay problem. What the present occasion supplies is a reason to state them on their own account — an announced solution to the official formulation — together with one argument that chapter had no occasion to make, the viscosity test of Section 6. A reader who knows that chapter will find what follows familiar, and should.

We propose two things, and it is as well to say at the outset that they are both ours and that neither is offered without the other. The first is a *reformulation*: a statement of the Navier–Stokes problem which asks about the flows that occur, at high Reynolds number, for data a fluid actually has. The second is a *solution of the reformulated problem*: not a proposal for how one might be found, but one given here, by construction, for specific data, with the error in what is claimed from it bounded rather than estimated. A reformulation without a solution would be an evasion, and a solution to a problem no one had stated would be of no interest. A reformulation that comes with its solution makes sense: it can be judged twice over, on whether the problem is the right one and on whether the answer is right, and the two verdicts are independent. A reader may accept the reformulation and dispute the solution, or accept the solution and think the official problem was fine as it stood. Both are available, which is more than either half offered alone would allow.

Proposing the two together is not a departure from mathematical practice. It is mathematical practice. A theorem and its proof are not found in the order in which they are printed: they are constructed in parallel, and the statement is adjusted as the argument develops until the two fit. Hypotheses are added because the proof needs them, conclusions are weakened because the proof cannot reach them, and what appears in the journal is the pair that closed. Nobody regards this as illegitimate, and it is not — but it means that a formulation *fixed in advance of any proof*, by a third party, and then left standing for a quarter of a century, is an unusual object, and one should

not be surprised if what eventually satisfies it is an argument built to fit rather than a discovery about fluids. The announced solution is the theorem–proof coupling working exactly as it always does, with the difference that the theorem was not the provers' to adjust, so the fitting had to be done entirely on the side of the construction — in this case by building the force (Section 5). We adjust the statement instead, and say so.

We take them in that order, and compare with the official formulation and the AI solution afterwards — the reformulation and its solution first, because a formulation is best judged against one that has a solution.

> **Problem.** *Constructive existence and qualitative uniqueness of weak solutions to the incompressible Navier–Stokes equations with small viscosity, that is at high Reynolds number. Given data:* a domain $\Omega \subset \mathbb{R}^3$, initial data u_0 on Ω, a forcing f on $\Omega \times (0, T)$, and boundary conditions on $\partial\Omega$.

Both adjectives carry weight, and what they exclude is as important as what they ask. Each is weaker than the classical demand in one respect and stronger in another, and in both the trade is the same: less is asked of the pointwise field, more of what can be done with it.

Constructive existence, as against existence only. Classical existence is an assertion that the solution set is non-empty. Leray's theorem [27] is of that kind: obtained by compactness, it establishes that a weak solution exists and says nothing whatever about which one, or what it looks like, or how to get at it. For a problem whose whole difficulty is that the solutions are complicated, this is close to no information. We ask instead that existence be established *by exhibition*, and concretely this means that five things be produced and not merely asserted: the mesh the solution lives on, with its cell count and its smallest cell; the discretisation carrying it, here continuous piecewise linear in space and time with least-squares control of the residual; the residual itself, evaluated in the norm the error bound uses, as a measured number and not an assumption; the computed stability factor for each output asked of it; and then, for each such output — the drag of a body, the lift of a wing, a dissipation rate, a pressure distribution — a value with a bound on its error. That is what the solution *de facto is*. Ask a non-constructive existence theorem for the drag and there is nothing there to ask: it has established that the set of solutions is not empty and has produced no member of it.

There is more to this than error control, and it is the part we would press. A computed solution is an *elaborate explicit construction of a velocity–pressure field* — some millions of numbers on a spacetime mesh, every one of them available — and such an object can be taken apart. One can ask it where vorticity concentrates and in what shape; how the velocity gradient decomposes at a point into straining, rotation and shear; where energy passes from large scales to small and through what structures; where the dissipation actually sits, as against where a model would put it. None of these questions can be put to a solution known only to exist. They can all be put to a computed one, and answered by inspection rather than by assumption.

That is not merely convenient. It is how the analytical questions get found. The triple decomposition of the velocity gradient [16], and the energy stability analysis of turbulent flow built on it [17], came out of looking closely at resolved velocity fields, and at measured ones [8]. An analytical statement about turbulence was prompted by inspecting constructed fields — which is the ordinary route by which computation has served analysis wherever both are practised, and the route a non-constructive existence theorem cannot offer.

We would put the claim this way. A solution concept whose objects can be inspected generates conjectures; one whose objects cannot be exhibited does not. If the difficulty of the Navier–Stokes

equations is that their solutions are complicated, then a formulation which asks for solutions one can examine is asking for the thing most likely to make the difficulty tractable.

It is worth noting that this is the sense in which the announced solution [22] is also constructive, and that the direction is what differs. It constructs, to order, a force that produces a chosen singularity; we construct, from data given in advance, a flow and the bound on what may be asked of it. Both exhibit. Only one exhibits a fluid.

Qualitative uniqueness, as against uniqueness only. Classical uniqueness asks that the data determine the field pointwise. At high Reynolds number this is false, and not marginally: two computations of the same flow on two meshes give velocity fields that differ everywhere and continue to differ under refinement. To ask for pointwise uniqueness is to ask for something the equations do not deliver, and a formulation that asks it cannot be satisfied by the solutions that occur.

What the data *do* determine is the qualitative content, and concretely this is what two computations of the same flow on independently generated meshes are found to share: the drag agrees to a few per cent [33] and the lift likewise; the separation line sits in the same place; the wake is organised the same way and sheds at the same frequency; the total dissipation rate is the same. What they do not share is the velocity, which at any chosen point differs by order one and goes on differing as the meshes are refined.

Qualitative uniqueness is the demand that this be so, and that the line between the two lists be drawn *quantitatively* rather than by assertion. That last clause is what keeps it from being a soft version of uniqueness. The adjoint solution attaches to each functional a computed stability factor: where it is moderate the output is determined by the data, to a precision the factor states, and refinement of the mesh where the adjoint is large will improve it; where the factor is large the output is not determined, and no refinement will make it so. So the claim is falsifiable one functional at a time, and a functional can fail it (Sections 2 and 1.1).

Three further things distinguish this from the official formulation.

The domain is given, not $\mathbb{R}^3$ with decay. Real flows occupy bounded domains with boundaries, and the boundary is where drag and lift are produced. A formulation on all of $\mathbb{R}^3$ removes the only place the physically interesting outputs live.

Small viscosity is part of the problem, not a parameter to be quantified away. The official statement fixes ν and asks a question about $t \to \infty$; here $\nu \to 0$ is the limit of interest and T is finite. This is the difference between asking about one Reynolds number at a time and asking about turbulence.

Weak solutions are asked for, and smoothness is not. Existence of weak solutions has been available since Leray [27], and by the Onsager theorem [25, 29] non-smoothness is not a defect to be removed but the mechanism by which a turbulent flow dissipates energy at a rate independent of viscosity. Requiring smoothness excludes the solutions that occur.

1.1 The same demand stated in outputs

The uniqueness half of that demand can be sharpened, and it is worth stating in the form in which it is actually answered — as a requirement on functionals, uniformly in the viscosity.

> **Problem.** Fix smooth divergence-free initial data of size one, and let $\nu > 0$ be arbitrary with no positive lower bound. Show that the weak solution is *wellposed in mean-value outputs, uniformly in ν*: that functionals of the form (1) depend continuously on the data with quantitative error control, by bounds that do not degenerate as $\nu \to 0$, even

though the pointwise solution does not and its gradients diverge as $\nu^{-1/2}$. Equivalently: establish the second law of thermodynamics for fluids — that a turbulent solution dissipates kinetic energy at a rate which does not vanish with the viscosity — the dissipation anomaly, or zeroth law, observed since Taylor [10] and still under active examination [9] — and that this dissipation is what makes the outputs stable.

Three features of this statement are worth marking. The *data* are normalised rather than the viscosity, which is the coordinate choice that keeps ν as a parameter and so allows a limit to be taken at all. The bounds are required *uniformly* in ν, which is what makes it a question about turbulence rather than about one Reynolds number at a time. And it asks for stability rather than regularity, so it can be answered affirmatively for the solutions that occur instead of being answered vacuously for solutions that do not.

2 Solution of the reformulated problem

The corresponding notion of solution is then also elementary, and it is the one the AI construction of Section 5 uses in a different direction: *a solution is a construction for specified data.*

Solution. *Computed solutions, as a form of constructed solution, for a representative choice of specific data — constructive existence by exhibition, qualitative uniqueness by the adjoint.* For a given Ω, u_0, f and boundary conditions, exhibit a computed weak solution together with quantitative control of the outputs: a residual that is small in the relevant norm, an error bound for each output obtained by duality from a computed solution of the adjoint equations, and a demonstration that the output converges under refinement while the velocity field does not.

The two constructions are worth setting side by side, since they are formally the same kind of object and opposite in intent. The AI construction fixes the singular solution wanted and solves backwards for a force that produces it, the force being whatever the residual requires; the resulting flow exists for the formulation and nowhere else, and vanishes as $\nu \to 0$ (Section 6). A computed solution fixes the data that a flow actually has — a body in a channel, a given inflow — and solves forwards, the residual being what is left over and what is then used to bound the error in the output. Both are constructions, both are specific, and only the second is about a fluid.

What makes the computed object a solution rather than an approximation is that the error in the output is controlled rather than hoped for. The adjoint equations turn the residual into a bound on $|M(\hat{u}) - M(\hat{w})|$ for the functional of interest, with computed stability factors; where those factors are moderate the output is predictable and the mesh can be refined where it pays; where they are large the output is not predictable and no amount of refinement will make it so. That is a statement about which questions the equations answer, and it is available without smoothness, which is why it survives at the viscosities where smoothness does not.

3 Output wellposedness, and why it is the right demand

The defect in the official formulation is not that it asks a hard question. It is that it asks about the wrong property. Smoothness is not what makes a solution concept physically meaningful. Wellposedness is.

Hadamard's criterion has three parts: a solution exists, it is unique, and it depends continuously on the data. The third is the one that carries the physical content, because data are never exact.

A problem whose solution is not stable under perturbation of the data is not a model of anything measurable, whatever its smoothness.

Weak solutions exist. Leray showed in 1934 that weak solutions of (10) exist globally in time [27]. For *small* data — which for suitably scaled problems is the low Reynolds number case — global existence of a *smooth* solution has been known for sixty years [1, 2], so the difficulty lies entirely at the other end, which is the end the reformulation addresses. What was left open was not existence but whether the weak solution concept is *wellposed* — whether it determines anything. That is the question worth asking, and it is not the question the Clay problem asks.

Wellposedness is a property of outputs, not of solutions. Here is the point on which the reformulation turns. For a turbulent solution, the pointwise velocity at a given place and time is not determined: perturb the data in the last decimal and the pointwise field is different everywhere within a short time. In that sense the solution is not wellposed and never will be.

But nobody measures the pointwise velocity of a turbulent flow, and no engineering question depends on it. What is measured, and what is asked for, are *mean-value outputs*: the drag on a body, the lift on a wing, a pressure averaged over a surface, a flux through a section. These are functionals of the solution, and the question of wellposedness must be asked of them:

$$M(u) = \int_Q u \cdot \psi \, dx \, dt \tag{1}$$

for a given weight ψ. The claim is that $M(u)$ is stable for turbulent solutions even though u is not.

The dual solution is a weight function connecting residual to output. Output quality is not assessed by inspection; it is computed. The error in an output is, to leading order, the residual of the computed solution tested against a *dual weight function* φ,

$$M(\hat{u}) - M(\hat{U}) \;\approx\; \int_0^T \int_\Omega R(\hat{U}) \cdot \varphi \, dx \, dt, \tag{2}$$

so φ is precisely what converts a residual — which one has, and can measure — into an error in the quantity one cares about. Everything in the assessment of an output is carried by φ: where it is large the residual there matters and the mesh must be refined; where it is small the residual is harmless however large; and its size overall is the stability factor that says whether the output is determined at all.

And φ is obtained by solving a definite problem: the Navier–Stokes equations linearised about the computed solution, run *backward in time* from the output functional as data at $t = T$. This is the same object, and the same computation, as the backward pass in the training of a neural network — the adjoint of a forward evolution, propagating sensitivity from an output to everything that fed it. Fluid mechanics and machine learning are doing the same linear algebra in opposite directions through the same trajectory.

Why the backward problem is stable for a turbulent solution. That the dual problem can be solved at all is the crux, and it is where intuition points the wrong way. A linearisation about a chaotic trajectory has exponentially growing modes; run backward, one expects φ to blow up and the whole assessment to be vacuous. It does not, and the reason is the oscillation.

A turbulent solution does not grow exponentially; it *oscillates*, and it oscillates precisely *because* it is the result of exponential growth followed by exponential decay — a structure is stretched

and amplified, becomes unstable, breaks down and dissipates, and the next takes its place. The linearised coefficients along such a trajectory therefore change sign and structure repeatedly, and the growth the dual solution accumulates over one interval is largely undone over the next. What would be a monotone exponential over a smooth field becomes, over a turbulent one, a product of factors alternately greater and less than one, and φ grows only modestly. Over a *smooth* field there is nothing to cancel, and it grows.

So turbulence stabilises the outputs that smoothness destabilises, and it does so by the same mechanism that makes it unpredictable pointwise. The growth and decay which destroy the velocity as a predictable quantity are what keep the weight function bounded and the mean value predictable.

This is the sense in which the reformulated problem has an answer, and the answer is not the one the dichotomy anticipates. Smooth solutions are not wellposed. Turbulent solutions are wellposed in mean-value outputs. The physically meaningful solution concept is the one the official formulation treats as the failure case.

4 In what sense this constitutes a solution

The solutions are constructed, not assumed. A finite element method for (10) with residual-based stabilisation — Galerkin least squares on a mesh [15], piecewise polynomial velocity and pressure, no turbulence model supplied by the user (§6) — produces solutions which become turbulent even when initiated from smooth potential data, and which stay bounded. The stabilisation supplies dissipation in proportion to the residual, so the dissipation is generated by the computation rather than prescribed by a model. In the vanishing-viscosity limit the same scheme is an Euler solver whose turbulent dissipation appears automatically. The non-smooth, non-singular third case is therefore not a conjecture: it is what the equations produce when they are solved.

The residual behaves as the theory requires. For such solutions the Navier–Stokes residual $R(U,P)$ is small in a weak norm and large in a strong one. Small weakly is what makes the computed field a solution; large strongly is the statement that it is not smooth. A method whose residual were small in both would be computing something that does not exist.

The error is controlled by duality, not by asserted convergence. An a posteriori error estimate of the form

$$|M(u) - M(U)| \;\le\; \int_Q R(U,P)\cdot\varphi\,dx\,dt \;\le\; S\,\|R(U,P)\|_{-1}, \tag{3}$$

with φ the dual solution and S its stability factor, converts a computed solution into a statement with a bound attached. The stability factor is not assumed; it is computed by solving the dual problem. It is through (3) that the wellposedness claim becomes quantitative: S is moderate for mean-value outputs over turbulent solutions and large for pointwise outputs and for smooth solutions, which is the distinction of §3 expressed as a number.

Computation as proof. That an answer rests on computation is not a defect peculiar to this problem. The four-colour theorem was settled the same way, and the settlement was accepted because the computation was reproducible and its error controlled. The same standard applies here, and (3) is what meets it: the claim is not that a particular simulation looked turbulent, but that the output of the computation comes with a bound which can be evaluated, and which is small.

5 Test of the OpenAI solution against the reformulation

On 8 September 2026 OpenAI announced a solution of the Clay problem in the form of a proof that a singular solution to the Navier–Stokes equations exists [22], obtained with ten thousand agents and of the order of 3×10^{11} output tokens, and the announcement drew attention as much for how the proof strategy was arrived at as for the result [23]. If it stands the scrutiny of the mathematics community, the Clay problem as posed is solved.

What it is a solution *of* deserves stating precisely, and the paper is explicit. Theorem 1.1 asserts that for every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty))$, a compact set K, and smooth fields u, p on $\mathbb{R}^3 \times [0,1)$ solving (10) with $u(\cdot,0) = 0$, supported in K, with $\sup_t \|u(t)\|_{L^2} < \infty$ and $\limsup_{t\uparrow 1} \|u(t)\|_{L^\infty} = \infty$. The flow starts from rest; everything in it comes from the force.

The order of construction is the point, and it is stated in the paper's own words. The velocity and pressure are built first, to have the required growth, and the force is then defined as what is left over: the aim in the final section is to give the local field "zero initial datum and compact spatial support, while making its momentum residual the restriction of a force in $C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$", achieved by proving that every derivative of the residual has a uniform limit as $t \uparrow 1$ and then constructing a force attaining those limits. An exponentially small force also seeds each of the oscillatory pulses, the background shear supplying their growth. So the force is not a physical forcing mechanism but the residual of a flow chosen in advance, arranged to be smooth; the singularity is supplied to the equations rather than generated by them, and the admissible class is broad enough to let it be supplied. That the class is what carries the result is already visible from the other side: under the structural features of the announced solution, replacing the C^∞ force by a spatially real-analytic one rules the mechanism out [24].

Two consequences follow, and this paper is about the second.

The first is that the official question has an answer, and the answer is the singular branch of the dichotomy. Nothing below disputes that. It is worth noting that the paper states its own scope in just these terms: "This establishes alternative (C) in the Millennium problem statement for Navier–Stokes as stated by Fefferman", with the periodic case giving (D). It does not claim (A) or (B), and on the reading argued in Section 8 it could not, since those are the unforced alternatives. We add only the observation, made quantitative in Section 6, that such a construction does not survive the small-viscosity limit in which fluid mechanics lives: held to a fixed admissible forcing class it shrinks to the zero solution, and held at fixed amplitude it requires a force growing like ν^{-1} that leaves the class. The singular point solution vanishes as $\nu \to 0$. It is a solution of the problem, not of the flow.

The second is that the implication for fluids is unclear, and that the unclarity is not repaired by strengthening the proof. Between the smooth branch and the singular branch sits the case that occurs — bounded velocity, gradients growing like $\nu^{-1/2}$, dissipation that does not vanish with the viscosity — and the dichotomy contains no room for it. This is the burden of the next section, and it was the burden before the announcement.

6 The viscosity test applied to the announced construction

The test is general: it applies to any construction that exhibits a singularity for Navier–Stokes at prescribed forcing, and in particular to the one described in Section 5. Its conclusion is that the singular point solution *vanishes* in the limit that fluid mechanics occupies.

Navier–Stokes rescales. If v solves the equations at viscosity one with forcing g, then

$$u(x,t) = a\, v(bx, ct), \qquad a = \nu b, \quad c = ab, \quad f = a^2 b\, g, \tag{4}$$

solves them at viscosity ν. The map is onto: *one* unit-viscosity blowup generates the entire family. A theorem asserting blowup "for every $\nu > 0$" is therefore not a family of results. It is one result, rescaled, and it contains exactly what its $\nu = 1$ member contains.

Everything then turns on what (4) does to the constants. Since it is the forcing class that Fefferman's formulation fixes, two normalisations bear on it, and the exponent quoted must always be accompanied by the one it belongs to. Hold the solution at size one, $a = 1$: then $b = c = \nu^{-1}$ and the forcing amplitude is $a^2 b = \nu^{-1}$, so the force required *blows up* as the viscosity falls and leaves any fixed admissible class — including the one Fefferman specifies, which fixes the constants $C_{\alpha K}$ once and for all. Or hold the forcing in a fixed class, $a^2 b = 1$: then $b = \nu^{-2/3}$ and

$$\text{amplitude } a = \nu^{1/3} \to 0, \qquad \text{length } b^{-1} = \nu^{2/3} \to 0, \qquad \text{time } c^{-1} = \nu^{1/3} \to 0. \tag{5}$$

The singularity shrinks to nothing: at $\nu = 10^{-6}$ the velocity reaches 10^{-2} on a scale of 10^{-4} before blowing up. Neither branch survives the limit. One exits the admissible data, the other converges to the zero solution, and there is no third option.

Holding the blow-up time, and the critical path. The freedom in (4) is a single parameter: given ν, a choice of b fixes $a = \nu b$ and $c = \nu b^2$. Two further ways of spending it are worth reporting, because they locate the construction more sharply than either branch above.

Hold the blow-up time. If v becomes singular at $s = 1$ then u does so at $t = 1/c$, and requiring $t = 1$ gives $b = \nu^{-1/2}$, whence

$$\text{amplitude } a = \nu^{1/2}, \qquad \text{length } b^{-1} = \nu^{1/2}, \qquad \text{forcing } a^2 b = \nu^{1/2}. \tag{6}$$

With the singular time pinned at $t = 1$, every constant in the construction vanishes like $\nu^{1/2}$ together.

The critical path. Keep that normalisation and ask where the velocity is of order one. Self-similar Navier–Stokes blow-up carries $|v(s)| \sim (1-s)^{-1/2}$ on length $(1-s)^{1/2}$, so

$$|u(t)| \;\sim\; \frac{\nu^{1/2}}{(1-t)^{1/2}} \;=\; \Big(\frac{\nu}{1-t}\Big)^{1/2}, \tag{7}$$

which reaches order one along

$$1 - t \sim \nu \tag{8}$$

and nowhere earlier. On that path the active length is $b^{-1}(1-t)^{1/2} \sim \nu^{1/2} \cdot \nu^{1/2} = \nu$, so the local Reynolds number is

$$\mathrm{Re}_{\mathrm{loc}} = \frac{|u|\,\ell}{\nu} = \frac{1 \cdot \nu}{\nu} = 1 \tag{9}$$

identically along it. An active length ν lies *below* the Kolmogorov scale $\nu^{3/4}$. So wherever this construction carries a velocity of order one it is a viscous-scale event beneath the dissipation scale at unit local Reynolds number — locally laminar, and in no part of the regime the problem advertises. This is a stronger statement than the vanishing of the family: it says that even where the construction is not small, it is not turbulent.

This is the sense in which the announced solution [22] settles the official question and leaves the physical one untouched. Its force is built from the residual to match a chosen singular solution (Section 5); the scaling then fixes what happens to that solution as the viscosity falls. A singularity that exists at every fixed ν and disappears as $\nu \to 0$ is not a feature of high-Reynolds-number

flow; it is a feature of the formulation, which admits forces no mechanism supplies. The independent exclusion of the mechanism under analytic forcing [24] points the same way from a different direction.

Set this beside the flow the problem is advertised as being about. Turbulence at small viscosity has *bounded* velocity at every scale and gradients growing like $\nu^{-1/2}$, realised by (12) on structures of size $\nu^{3/4}$ across which the velocity varies by $\nu^{1/4}$ — a Hölder-1/3 field, amplitude fixed, structure refining. The constructed family has amplitude vanishing like $\nu^{1/3}$, which is the opposite behaviour, or else forcing diverging like ν^{-1}, which is not a fluid at all.

7 Cases of computed solutions, with assessment of uniqueness

This section is the answer to the reformulation of Section 1, in the only form such a question can be answered: specific data, a computed weak solution, and an assessment of what is unique about it. The method, the error control and the computations are set out in *Computational Turbulent Incompressible Flow* [30], with the residual-based dissipation in [18], the parameter-free adaptive formulation in [19], the earlier statement of the approach in [21], and a case at aeronautical Reynolds number — the DLR-F11 aircraft model, time-resolved and adaptively refined — in [20]. Further material is collected under the label *clay problem* [35].

The pattern across those cases is the one the formulation predicts, and it is worth stating before the details. In each, the velocity field fails to converge under refinement and is not expected to: two meshes give two different fields, and the difference does not diminish. In each, the mean-value output converges to a few per cent, and the error in it is *bounded*, not estimated after the fact, by duality against a computed adjoint solution. And in each, the adjoint supplies the assessment of uniqueness directly: where its stability factors are moderate the output is unique in the sense that matters and the mesh is refined where that pays; where they are large the output is not predictable and refinement will not make it so. That last case is not a failure of the method but an answer — the equations do not determine that quantity, and no computation should be believed which claims otherwise.

G2: the method is Galerkin, and the turbulence model is the residual. General Galerkin, or G2, is a stabilised finite element method — continuous piecewise linear in space and time, cG(1)cG(1), with least-squares control of the residual. The stabilisation is not a subgrid model: no eddy viscosity is prescribed, no wall function, no closure, no constant fitted to measurement. The dissipation is whatever the residual makes it, which means it is generated by the flow being computed rather than supplied by the person computing it. This matters for the present argument specifically. A computation carrying a fitted turbulence model could not be used to say what Navier–Stokes contains, because the turbulence would have been put in by hand. G2 has nothing put in by hand, so what comes out is a property of (10).

The dual problem is solved, and the stability factor is a computed number. The estimate (3) is the book's organising device. For each output of interest the dual problem is linearised about the computed solution, solved backwards, and its stability factor evaluated; the mesh is then refined where the product of residual and dual weight is largest. The consequence is that the error in the output is not estimated by comparing successive meshes and hoping, but bounded by a quantity the computation itself produces. Adaptivity and error control are the same mechanism.

Mean-value outputs converge; the flow field does not. The book's central computational finding is the one §3 needs. Mean drag and mean lift for turbulent flow past a bluff body come out to a tolerance of a few per cent on meshes of a few hundred thousand points — small enough to run on a personal computer, and orders of magnitude coarser than a direct simulation resolving the Kolmogorov scale would require. The pointwise velocity field on those meshes is not converged and does not become so. Both facts hold simultaneously, and together they are the wellposedness claim made quantitative: the outputs are stable because the stability factor for them is moderate, and the pointwise field is not because its stability factor is not.

Turbulent Euler solutions and d'Alembert's paradox. Taking $\nu \to 0$ in G2 gives a solver for the Euler equations whose dissipation does not vanish with the viscosity, which is the second law of §4 in computational form. What such solutions produce is not potential flow. A bluff body in this limit shows potential flow modified by a three-dimensional rotational separation, and that separation carries drag — which is the resolution of d'Alembert's paradox offered in [31]: the paradox arises from taking the smooth potential solution to be the vanishing-viscosity limit, and the correct limit is turbulent. The same analysis applied to a wing gives lift and drag without a circulation condition imposed by hand [32].

This is the concrete content behind the claim of §2 that smooth solutions are unstable. It is not an assertion about what would happen to a perturbed smooth solution; it is a record of what happens when the equations are solved from smooth data without a model to keep them smooth.

8 The official formulation recalled

The Clay Mathematics Institute problem, as set out by Fefferman [26], concerns the incompressible Navier–Stokes equations

$$\dot{u} + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \tag{10}$$

on $\mathbb{R}^3$ or a periodic box, with $u(\cdot, 0) = u^0$ smooth and divergence free, and with the viscosity ν normalised to one. Four statements are offered, of which a proof of any one settles the prize: existence and smoothness on $\mathbb{R}^3$; the same with periodic data; or a counterexample to either — a solution which is initially smooth and whose velocity becomes unbounded in finite time.

The structure of the question is therefore a dichotomy. Either the solution is smooth for all time, or it is not, and the only stated way for it not to be is blowup. *Tertium non datur.*

9 Is the alternative exhaustive?

The physical solution is neither. A fluid at a Reynolds number of interest — 10^6 and upward for an aircraft, a ship, a weather system, a pipe — is turbulent. Turbulent flow is not smooth in any useful sense: the velocity varies on a length scale that shrinks with viscosity, and the gradients scale as

$$|\nabla u| \sim \nu^{-1/2}, \tag{11}$$

so that at $\nu = 10^{-6}$ they are of order 10^3 times the velocity scale, and at the viscosity of air over a wing, larger still. Nor is it singular: the velocity stays bounded, the kinetic energy stays finite, and nothing goes to infinity anywhere. Kolmogorov's scaling [3, 4], in the form Onsager gave it, puts the velocity field at Hölder continuity $1/3$ — refined by the intermittency corrections of [5, 6] and the multifractal picture of [7], and continuous, with unbounded derivatives in the limit. That is precisely the case the official alternative does not contain.

The smooth case is trivial or false. If smoothness is to be read as "the solution exists and is smooth for a short time", the statement is classical and was not worth a prize. If it is to be read as "the solution stays smooth for all time", then for the flows of interest it is false, and not because of blowup: a smooth solution of (10) at small viscosity is *unstable*. Perturb potential flow around a bluff body and it does not stay potential flow; it develops rotational separation and becomes turbulent. The smooth solution exists as a formal object and is not observed, because nothing keeps a flow in it. So the first horn of the dichotomy describes solutions that are mathematically available and physically absent, and the second describes solutions that do not occur.

Turbulence is a necessity, not an accident. The instability just invoked is not a vague appeal to perturbations; it has a mechanism, and the mechanism ends in turbulence because nothing else is available to it.

A high Reynolds number flow set going laminar develops *shear instabilities*. A velocity profile with a sufficiently sharp transverse gradient is unstable, and the instability does not relax the gradient but steepens it: the shear layer rolls up, and the rolled-up structure presents steeper shear still. Steepening continues until the gradient is sharp enough for viscosity to act on it, and at small ν that means very sharp indeed — gradients of order $\nu^{-1/2}$ by (11), on structures of order $\nu^{3/4}$. Only at that scale is there dissipation to balance the energy supplied from the large scales. A state with gradients steep enough to dissipate and a velocity which nevertheless stays bounded is exactly what turbulence is.

The consequence is stronger than "the laminar solution is not observed". It is that the laminar solution has no *continuation*. It cannot persist, because the instability grows. It cannot escape by blowing up, because the steepening is arrested the moment viscosity bites. And it cannot dissipate in the laminar state what the large scales feed it, because laminar gradients at high Reynolds number are far too small to dissipate at that rate. One route remains, and the flow takes it: it becomes turbulent *in order to continue to exist at all*. Every high Reynolds number laminar flow is in this position, so every one of them turns turbulent.

This is why the case the official formulation treats as failure is the case that has to occur. Turbulence is not a pathology of the equations, nor a loss of the solution: it is the mode in which solutions exist at high Reynolds number, and non-smoothness is not the damage but the mechanism — by the Onsager theorem [25, 29] it is precisely what allows dissipation at a rate which does not vanish with the viscosity. A formulation which admits only "smooth for all time" and "velocity unbounded" has excluded the only continuation the equations possess.

The normalisation is legitimate; what is wrong is the quantifier. It might be thought that setting $\nu = 1$ removes the phenomenon, since all the structure in (11) is structure in ν. Taken as an objection to the normalisation itself, that is wrong. Navier–Stokes rescales as

$$v(x,t) = \alpha\, u(\beta x, \alpha\beta t), \qquad \nu' = \nu\beta/\alpha,$$

so any viscosity can be set to one by rescaling the data: small ν with data of size one is the *same problem* as $\nu = 1$ with large data. Since Fefferman quantifies over all Schwartz-class data, arbitrarily high Reynolds number is inside the hypotheses, and the family of problems is the right one.

The defect is what is asked of that family. Fefferman asks, *for each* datum separately, whether a global smooth solution exists. Turbulence is not a property of any one member; it is a property of the family as the Reynolds number grows, and a question about it must be asked *uniformly*. Put the family in its natural coordinates — data of size one, $\nu > 0$ with no positive lower bound — and the distinction is immediate:

quantity	bounded for each fixed $\nu > 0$?	bounded uniformly as $\nu \to 0$?
velocity $\|u\|$	yes	**yes** (energy)
gradient $\|\nabla u\|$	yes	**no** — grows as $\nu^{-1/2}$
mean-value output $M(u)$	yes	**yes** — and this is the content

Read down the first column and every answer is yes, which is Fefferman's question and which settles nothing. Read down the second and the three quantities separate, which is the question about fluids. A theorem of the form "for each ν the solution is smooth, with derivatives bounded by an iterated exponential" is true in the first column and empty in the second, since the bound degenerates precisely in the limit at issue. That is the vacuity, and it survives the scaling argument rather than depending on it.

Direct simulation settles the matter by exhibition. The argument need not stay abstract, because solutions of the kind at issue are computed routinely. Direct numerical simulation resolves the flow down to the Kolmogorov cutoff $\eta \sim \nu^{3/4}$, which requires $L/\eta \sim \mathrm{Re}^{3/4}$ points per direction and $\mathrm{Re}^{9/4}$ in all — a thousand points a side at $\mathrm{Re} = 10^4$, which is ordinary. The field so computed is *smooth by construction*: it is fully resolved, every derivative is finite, nothing is singular anywhere, and no structure is left below the mesh.

It is also, on inspection, turbulent. One can look at it.

So here are solutions which satisfy the first horn of the dichotomy exactly — smooth, existing for all time, no blowup — and which are the very phenomenon the problem is supposed to be about. If (A) were proved tomorrow it would be affirmed by these fields, and would say nothing whatever about them that distinguishes them from laminar flow. The predicate does not separate the cases it is being asked to separate. That is not an objection to the proof techniques; it is an objection to the question, and direct simulation makes it not an argument but an observation.

The real gap, and the question it should have become. It is worth being exact about what *is* open here, because there is a genuine gap and the Clay problem was posed in its vicinity. That turbulent flow has a smallest scale set by the viscosity is doubted by almost nobody, and on very substantial grounds: dimensional analysis, the observed independence of the dissipation rate from ν, and a century of measurement. But it is not a theorem about (10). Kolmogorov's $\eta \sim \nu^{3/4}$ rests on local isotropy and universality assumed, not derived; the zeroth law is an empirical regularity; and that Navier–Stokes solutions converge as $\nu \to 0$ to anything in particular is unproved. A mathematical question is genuinely wanted here, and that is the impulse behind the prize.

What is proved lies on the Euler side and is worth setting against the dichotomy. Onsager's threshold is settled in both directions: above Hölder $1/3$ the energy is conserved [28], and below it dissipating solutions exist and have been constructed [29], after the partial results of [11, 12, 13]. So the third case — bounded, not smooth, dissipating — is not a conjecture about what solutions might do. It is a category with occupants.

The question the gap asks is therefore: *does the cascade terminate at a scale set by the viscosity, and does the field stay bounded while its gradients do not?* That is a question about the structure of the solution, and either answer is informative. Fefferman's formulation asks instead whether the velocity stays finite. The two are not the same question, and the second cannot express the first: a solution whose cascade fails to terminate is non-smooth without being singular, and the dichotomy has no name for it. The inspiration was sound and the formalisation lost it.

The quantitative results confirm rather than relieve this. Work in the analytic tradition has produced conditional regularity bounds of the form: if the velocity is bounded by A, then the

derivatives are bounded by an expression exponential in A iterated several times. As a theorem this is real. As a statement about a fluid it is empty: a bound of the size $\exp\exp\exp A$ is not a bound that any flow can be said to respect, and no measurement could distinguish it from no bound at all. The objection sometimes made to the problem's framing — that the selected question has little physical content, and reads as one posed from harmonic analysis rather than from continuum mechanics — is, on this evidence, well taken.

Non-smooth and non-singular, quantitatively. The two halves of the third case are one scaling seen from either end. Dissipation at a rate independent of the viscosity, $\varepsilon \sim \nu|\nabla u|^2$ held fixed as $\nu \to 0$, gives (11); and the scales at which that gradient is realised are fixed by the same constant ε:

$$\eta \sim \left(\frac{\nu^3}{\varepsilon}\right)^{1/4} \sim \nu^{3/4}, \qquad \delta u(\eta) \sim (\varepsilon\,\eta)^{1/3} \sim \nu^{1/4}, \qquad \frac{\delta u(\eta)}{\eta} \sim \nu^{-1/2}, \tag{12}$$

the last recovering (11). The middle relation is Hölder continuity with exponent $1/3$ — $\delta u(\ell) \sim \ell^{1/3}$, the Onsager exponent — evaluated at the smallest scale present. At $\nu = 10^{-6}$ the finest structure is about 3×10^{-5} of the macroscopic scale and the velocity varies across it by about 3×10^{-2}; at $\nu = 10^{-12}$, 10^{-9} and 10^{-3}. As $\nu \to 0$ the structures refine without limit while the velocity stays bounded at every scale: the gradient diverges, so the solution is not smooth, and the velocity does not, so it is not singular. That is the whole of the third case, and (12) is the reason no amount of regularity theory reaches it — a Hölder-1/3 field is exactly the object that is neither differentiable nor unbounded.

It also disposes of a tempting misreading. A singular structure produced by letting $\nu \to 0$ is not a counterexample to (10) at any fixed viscosity, because by (12) the finest scale present at that viscosity is $\nu^{3/4}$ and finite, and the solution there is bounded. Such a structure is singular only in the limit, where the equation being solved is Euler's and not Navier–Stokes's. The Clay problem is posed at fixed positive viscosity. A construction whose singularity shrinks to nothing as the viscosity is removed has answered the Euler question — itself open, and interesting — and not the one asked. The same test applies to any claimed blowup: does the singular structure survive at fixed $\nu > 0$ with data satisfying the stated decay, or does it exist only in a limit, or only with a forcing term the formulation excludes?

What the problem advertises, and what it asks. The Clay Institute's own description of the problem opens on turbulence. Waves follow the boat across the lake, turbulent air currents follow the aircraft, and mathematicians and physicists believe an explanation of both the breeze and the turbulence is to be found in solutions of the Navier–Stokes equations; the challenge is to unlock the secrets hidden in them. That is the advertisement, and it is accurate about where the secrets are.

The formal statement that follows contains no turbulence. It asks whether the velocity stays finite. Between the description and the problem, the subject has been sorted out — not by oversight in the description, which names the phenomenon correctly, but by the formalisation, which admits no predicate that could distinguish a turbulent solution from a laminar one. The reader is promised the secret hidden in Navier–Stokes and handed a regularity question. That is the unfortunate part, and it is why this article argues for reformulation rather than for a different proof strategy.

The alternatives are not complementary, and the prize can be won without settling anything. Fefferman's four statements do not form two pairs of negations. (A) and (B) are

posed with $f \equiv 0$ and quantify over *all* admissible data: does every datum give a global smooth solution? (C) and (D) ask instead for the *existence* of admissible data *together with an admissible smooth forcing* f for which smoothness fails. So (C) is not ¬(A). A blowup driven by an external force leaves (A) untouched — it does not show that a fluid tears itself into a singularity, only that something can be done to it that does.

This is not hypothetical: the construction described in Section 5 is of exactly this kind, and discharges one of the four stated alternatives while leaving the question everyone means by the Navier–Stokes problem — can the equations, *unforced*, produce a singularity — precisely where it was.

The second thing to notice belongs to this article's subject rather than to the referees'. Such a construction gives, at each viscosity, a solution whose velocity becomes unbounded. Follow the family down as $\nu \to 0$ and the limit is singular. But that is not what happens to a fluid as the viscosity is reduced: what happens is turbulence, which by (11) and (12) has diverging gradients and bounded velocity, and by the Onsager results above is a category with occupants. A local energy equation can in fact be written in which the non-smoothness is the direct cause of the dissipation [14]. So the family constructed has no physical counterpart at either end — it is singular where fluids are turbulent — and the fact that it nonetheless answers one of the four stated alternatives is a fact about the alternatives. A formulation whose horns can be satisfied without bearing on its subject is a formulation in need of repair, and that is an observation about the statement, not about the proof.

The viscosity test. There is a test any such construction should be required to pass, and it is the one the formulation never asks for: *report the constants as functions of* ν. Navier–Stokes rescales, so a single unit-viscosity blow-up generates the entire family, and what distinguishes one construction from another is not that it blows up but how its forcing, amplitude and active length behave as the viscosity falls. Carried out in Section 6, the test gives, for each way of spending the rescaling freedom: a forcing that must diverge like ν^{-1} if the *amplitude* is held at one; an amplitude vanishing like $\nu^{1/2}$ if the *blow-up time* is held at $t = 1$; an amplitude vanishing like $\nu^{1/3}$ on support $\nu^{2/3}$ if the *forcing* is held in a fixed admissible class, which is the case that matters for Fefferman's statement since it is his class that is fixed; non-commuting limits, infinity one way and the zero solution the other; and, along the critical path $1 - t \sim \nu$ where the amplitude is of order one, a local Reynolds number identically equal to 1 and an active length ν, which is *below* the Kolmogorov scale $\nu^{3/4}$. The construction is a viscous-scale event taking place beneath the dissipation scale, and no part of it is in the regime the problem advertises.

(A) is provable, for an arbitrarily small change to the equation. There is a third way the dichotomy fails to bite, and it is the oldest. Augment the viscous stress by a term depending on a power of the local velocity gradient,

$$\dot{u} + (u \cdot \nabla)u - \nu\Delta u - \nabla \cdot \big(|\nabla u|^{\,p-2}\nabla u\big) + \nabla p = f, \tag{13}$$

so that the effective viscosity is $\nu + c|\nabla u|^{\,p-2}$ rather than ν. Then the standard machinery of monotone operators and compactness — Galerkin approximation, an energy estimate, a monotonicity argument to pass to the limit in the nonlinearity — delivers global existence and uniqueness of regular solutions. Lions [34] proved this in 1969 for

$$p \;\geq\; \frac{n+2}{2} \;=\; \frac{5}{2} \quad (n = 3), \tag{14}$$

that is, for viscosity augmented by $|\nabla u|^{1/2}$; later work lowered the threshold to $p \geq 11/5$, an augmentation by $|\nabla u|^{1/5}$. The exponent is small, and it can be multiplied by a coefficient c as small as one likes without affecting the theorem: the estimate needs the term's *presence*, not its size.

So Fefferman's alternative (A) — existence and smoothness — is not hard in itself. It is provable by techniques that were standard before the problem was posed, for an equation differing from (10) by a term with an arbitrarily small coefficient and an arbitrarily small power. What the prize turns on is the exact value $p = 2$ and not a hair more, which is a statement about a limiting case of a family whose every other member is settled.

Two things follow. The first is that the difficulty is concentrated at a point of no physical significance: nothing measurable distinguishes $c = 0$ from $c = 10^{-10}$, and the fluid does not know which equation it is supposed to satisfy. The second is sharper, and returns to the theme of this section. The added term $-\nabla \cdot (|\nabla u|^{p-2}\nabla u)$ is not a mathematical artifice introduced to make a proof work: with $p = 3$ it is the Smagorinsky model, and it is what turbulence modelling has been doing since 1963. The regularisation that makes (A) provable *is* a turbulence model. Fefferman's formulation excludes turbulence from its alternatives, and the price of that exclusion is precisely the term whose absence makes (A) hard — so the difficulty of the Clay problem is manufactured by the same decision that empties its dichotomy.

Turbulence removed, restored, and still not enough. The three observations combine into one. The problem is announced as a problem of turbulence and formulated as a problem without it: the description opens on the wake behind the boat and the air behind the aircraft, and the formal statement that follows asks only whether $|u|$ stays finite. Put the turbulence back in the one form analysis can digest — a turbulent viscosity depending on the local velocity gradients, as in (13) — and Fefferman's (A) ceases to be open. Pointwise solutions exist globally, no singularity forms, and the proof is the standard Sobolev-space argument of [34]: an a priori estimate, a compactness theorem, a monotonicity argument. The turbulence the formalisation discarded is precisely what makes the question answerable.

But it does not reach wellposedness, and this is where the matter rests. What (13) yields is a pointwise solution which exists and is unique. It does not yield a solution whose pointwise values mean anything. A turbulent flow is pointwise unstable whether or not a theorem asserts that its trajectory is unique: perturb the data in the twelfth digit and the velocity at a given point and time is different, while the drag, the lift and the mean pressure are not. Uniqueness in a function space and stability of an output are different properties, and only the second is what a computation, an experiment, or an aircraft depends on. A regularised equation therefore buys the prize's alternative (A) at the cost of answering a question nobody was asking, and leaves untouched the question everybody is: which outputs of this flow are stable, and to what tolerance.

That is the reformulation this article proposes, and the next section states it.

10 What the answer says about the original question

To the official question — do smooth data give solutions that remain smooth for all time — the answer implied by the above is no, and not by blowup. Smooth data give solutions that become turbulent: non-smooth, with gradients growing as $\nu^{-1/2}$, bounded in velocity, dissipating at a rate which does not vanish as $\nu \to 0$. Blowup, the only failure mode the official formulation admits, does not occur and is not the interesting question. The interesting question is the one the formulation leaves out.

This does not claim the prize, and it is not intended to. It claims something less procedural and more useful: that the problem as posed cannot be answered in a way that illuminates fluid mechanics, because the alternative it offers is not exhaustive over the cases that occur; and that the adjacent, answerable question — wellposedness of mean-value outputs of turbulent weak solutions — is both the physically meaningful one and one to which the answer is affirmative and computable.

A Conclusion: what the official alternative reaches, and what it does not

Put the two halves together.

The official statement offers four alternatives, and its (A) and (B) are posed with $f \equiv 0$ and quantified over *all* admissible data, while its (C) and (D) ask for the *existence* of data *together with an admissible forcing* for which smoothness fails. So (C) is not ¬(A), and the gap between them is exactly the force. A proof of (C) exhibits something that can be done *to* a fluid; (A) asks whether a fluid does it to itself. The announced solution [22] is a proof of the first kind, and its force is not a mechanism but a residual solved backwards (Section 5). *Even if it is formally correct, it does not settle* (A). The question everyone means by the Navier–Stokes problem is where it was on 7 September 2026.

And it does not settle anything about fluids either, for the reason of Section 6: held to an admissible forcing class the singular point solution vanishes as $\nu \to 0$, amplitude like $\nu^{1/3}$ on support like $\nu^{2/3}$, so the small-viscosity limit of the family is $u \equiv 0$; held at fixed amplitude the force diverges like ν^{-1} and leaves the class. A singularity that exists at every fixed viscosity and disappears in the limit where fluids become turbulent is a property of the admissible class, not of the flow.

So the verdict is symmetrical, and it is not a complaint about the proof. The formulation is mathematically sound — the theorem is a theorem, and the force is admissible — but it is *incomplete*: it admits a solution which decides neither of its unforced alternatives and no question about a fluid. The construction answers it exactly, and answers it by building a force to order. Neither is at fault given the other. What they close, between them, is a question about an admissible class rather than about turbulence.

What is about fluids is the reformulation of Section 1 and the solution of Section 2 — constructive existence in place of an assertion that the solution set is non-empty, and qualitative uniqueness in place of a pointwise determinacy the equations do not have: weak solutions at high Reynolds number, for a domain, initial data, forcing and boundary conditions that a flow actually has; uniqueness asked of mean-value outputs, which is true, rather than of the velocity field, which is false; and the answer given as a computed solution with the error in each output bounded by duality. That problem has a solution, the solution is turbulence, and the outputs are stable to a few per cent on meshes where the velocity field is not converged and never will be.

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